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Roll No. 3112413006

324741(24)

BE (7th Semester)

Examination, Nov:-Dec., 2016

(New Scheme)

Power Apparatus System

Time Allowed : 3 hours *Maximum Marks :* 80

Minimum Pass Marks : 28

Note : (*i*) Part (*a*) is compulsory in all questions. Attempt any **two** parts from (*b*), (*c*) and (*d*).

(*ii*) The figures in the right-hand margin indicate marks.

Unit-I

1. (*a*) Define strong efficiency. [2]

(*b*) On overhead transmission line at river crossing is supported from two towers at height of 40 m and 90 m above water level, the horizontal distance between the tower being 400 m. If the maximum allowable tension is 2000 kg, find the clearance between the conductor and water at a point mid-way between towers. Weight of conductor is 1 kg/m. [7]

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(*c*) Each line of 3-ϕ system is suspended by a string of 3 identical insulators of self capacitance C farads. The shunt capacitance of metal work of each insulator is 0.26 C to earth and 0.15 C to line, find string efficiency. [7]

(*d*) Explain in detail : [7]

(*i*) Suspension insulator

(*ii*) Any one method of improving string efficiency

Unit-II

2. (*a*) Define distributor system. [2]

(*b*) What are general consideration govern for design of distributor system? Bring out their relative importance. [7]

(*c*) Explain radial, ring main and inter-connected system in detail. [7]

(*d*) A 2-wire d-c street main AB 600 m long is fed from both end at 220V. Load of 20A, 40A, 50A and 30A are tapped at distance of 100m, 250m, and 400m and 500m from the end A respectively. If area of cross-section of distributor conductor is 1 cm^2, find minimum consumer voltage. Take $\rho = 1.7 \times 10^{-6} \Omega$ cm. [7]

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Unit-III

3. (*a*) What is the objective of Neutral grounding? [2]

(*b*) Explain arc suppression coil grounding in detail with proper diagram. [7]

(*c*) A 33kV , 3-ϕ, 50 Hz overhead line 60 km long has a capacitance to ground of each line equal to 0.015 r.f. per km. Determine inductance and kVA rating of the Peterson coil. [7]

(*d*) Explain equipment grounding in detail. [7]

Unit-IV

4. (*a*) Define overvoltage. Classify the cause of overvoltage. [2]

(*b*) Discuss :

(*i*) External and internal overvoltage

(*ii*) Mechanism of lightning discharge in detail [7]

(*c*) Explain insulation co-ordination in detail. [7]

(*d*) Explain in detail what are most commonly device used for protection of lightning surge. Also explain expulsion type arrester in detail. [7]

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Unit-V

5. (*a*) Define Reliability. [2]

(*b*) Explain bathtub curve in detail. [7]

(*c*) Derive an expression for probability density function. [7]

(*d*) The reliability of component is 0.3. How many such component be connected in parallel to achieve as over all reliability of at least 0.85? [7]

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